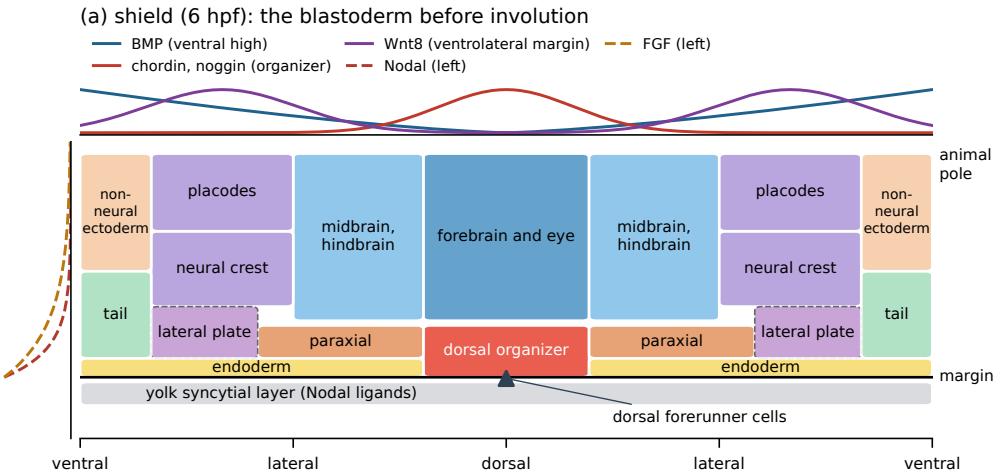
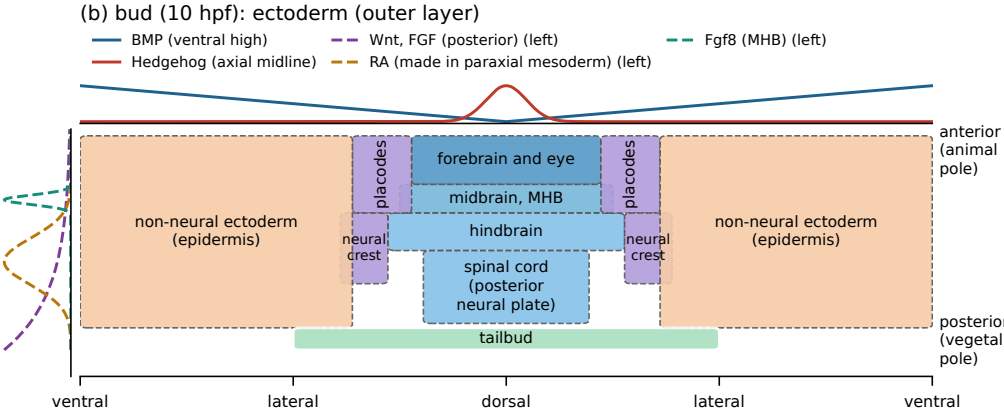
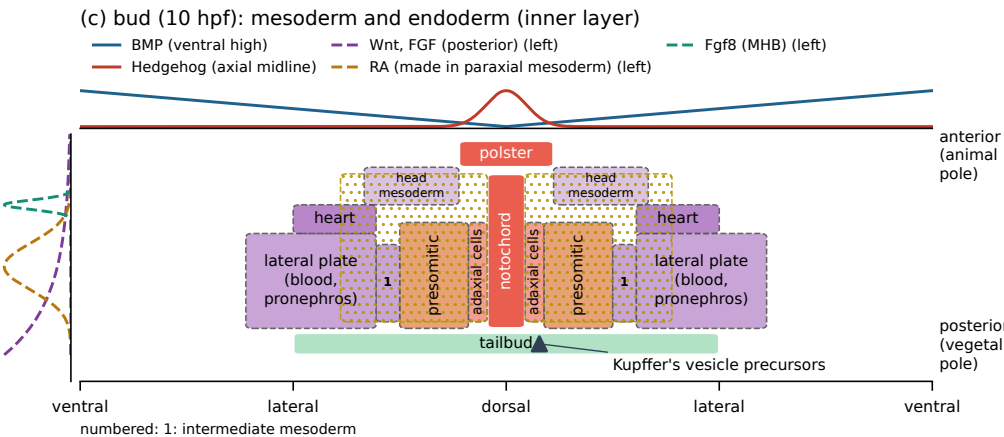

(a) shield (6 hpf): the blastoderm before involution
BMP (ventral high)
chordin, noggin (organizer)
Wnt8 (ventrolateral margin)
Nodal (left)
FGF (left)
non-neural ectoderm
placodes
neural crest
tail
lateral plate
midbrain, hindbrain
forebrain and eye
paraxial
dorsal organizer
endoderm
yolk syncytial layer (Nodal ligands)
dorsal forerunner cells
animal pole
margin
ventral
lateral
dorsal
lateral
ventral


(b) bud (10 hpf): ectoderm (outer layer)
BMP (ventral high)
Hedgehog (axial midline)
Wnt, FGF (posterior) (left)
RA (made in paraxial mesoderm) (left)
Fgf8 (MHB) (left)
non-neural ectoderm (epidermis)
placodes
neural crest
forebrain and eye
midbrain, MHB
hindbrain
spinal cord (posterior neural plate)
tailbud
anterior (animal pole)
posterior (vegetal pole)
ventral
lateral
dorsal
lateral
ventral


(c) bud (10 hpf): mesoderm and endoderm (inner layer)
BMP (ventral high)
Hedgehog (axial midline)
Wnt, FGF (posterior) (left)
RA (made in paraxial mesoderm) (left)
Fgf8 (MHB) (left)
polster
head mesoderm
heart
lateral plate (blood, pronephros)
1
presomitic
adaxial cells
notochord
tailbud
Kupffer's vesicle precursors
anterior (animal pole)
posterior (vegetal pole)
ventral
lateral
dorsal
lateral
ventral
numbered: 1: intermediate mesoderm